

GE Healthcare

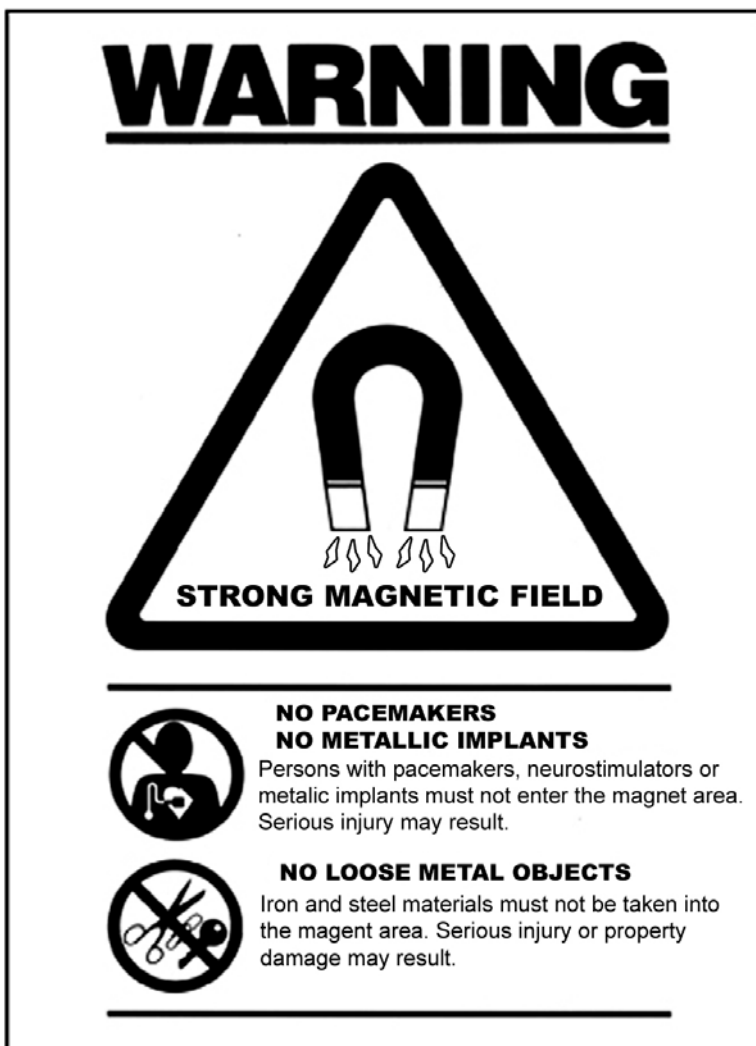
# High Order Shim Option

## Service Methods



OPERATING DOCUMENTATION

5314688  
Revision 1



# Important Information

## LANGUAGE

### ПРЕДУПРЕЖДЕН ИЕ (BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

### 警告 (ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

### VÝSTRAHA (CS)

- TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE.
- V PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKladU DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA.
- NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH.
- V PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ VLIVEM ELEKTRICKÉHO PROUDU, RESPEKTIVE VLIVEM MECHANICKÝCH ČI JINÝCH RIZIK.

**ADVARSEL****(DA)**

- DENNE SERVICE MANUAL FINDES KUN PÅ ENGELSK.
- HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE.
- FØRSØG IKKE AT SERVICE UDSTYRET MEDMINDRE DENNE SERVICE MANUAL HAR VÆRET KONSULTERET OG ER FORSTÅET.
- MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

**WAARSCHUWING****(NL)**

- DEZE ONDERHOUDSHANDLEIDING IS ENKEL IN HET ENGELS VERKRIJGBAAR.
- ALS HET ONDERHOUDSPERSONEEL EEN ANDERE TAAL VEREIST, DAN IS DE KLANT VERANTWOORDELIJK VOOR DE VERTALING ERVAN.
- PROBEER DE APPARATUUR NIET TE ONDERHOUDEN VOORDAT DEZE ONDERHOUDSHANDLEIDING WERD GERAADPLEEGD EN BEGREPEN IS.
- INDIEN DEZE WAARSCHUWING NIET WORDT OPGEVOLGD, ZOU HET ONDERHOUDSPERSONEEL, DE OPERATOR OF EEN PATIËNT GEWOND KUNNEN RAKEN ALS GEVOLG VAN EEN ELEKTRISCHE SCHOK, MECHANISCHE OF ANDERE GEVAREN.

**WARNING****(EN)**

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR, OR PATIENT FROM ELECTRIC SHOCK, OR FROM MECHANICAL OR OTHER HAZARDS.

**HOIATUS  
(ET)**

- KÄESOLEV TEENINDUSJUHEND ON SAADAVAL AINULT INGLISE KEELES.
- KUI KLIENDITEENINDUSE OSUTAJA NÕUAB JUHENDIT INGLISE KEELEST ERINEVAS KEELES, VASTUTAB KLIENT TÕLKETEENUSE OSUTAMISE EEST.
- ÄRGE ÜRITAGE SEADMEID TEENINDADA ENNE EELNEVALT KÄESOLEVA TEENINDUSJUHENDIGA TUTVUMIST JA SELLEST ARU SAAMIST.
- KÄESOLEVA HOIATUSE EIRAMINE VÕIB PÕHJUSTADA TEENUSEOSUTAJA, OPERAATORI VÕI PATSIENDI VIGASTAMIST ELEKTRILÖÖGI, MEHAANILISE VÕI MUU OHU TAGAJÄRJEL.

**VAROITUS  
(FI)**

- TÄMÄ HUOLTO-OHJE ON SAATAVILLA VAIN ENGLANNIKSI.
- JOS ASIAKKAAN HUOLTOHENKILÖSTÖ VAATII MUUTA KUIN ENGLANNINKIELISTÄ MATERIAALIA, TARVITTAVAN KÄÄNNÖKSEN HANKKIMINEN ON ASIAKKAAN VASTUULLA.
- ÄLÄ YRITÄ KORJATA LAITTEISTOA ENNEN KUIN OLET VARMASTI LUKENUT JA YMMÄRTÄNYT TÄMÄN HUOLTO-OHJEEN.
- MIKÄLI TÄTÄ VAROITUSTA EI NOUDATETA, SEURAUKSENA VOI OLLA HUOLTOHENKILÖSTÖN, LAITTEISTON KÄYTTÄJÄN TAI POTILAAN VAHINGOITTUMINEN SÄHKÖISKUN, MEKAANISEN VIAN TAI MUUN VAARATILANTEEN VUOKSI.

**ATTENTION  
(FR)**

- CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

**WARNUNG**

(DE)

- DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ**

(EL)

- ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ ΔΙΑΤΙΘΕΤΑΙ ΣΤΑ ΑΓΓΛΙΚΑ ΜΟΝΟ.
- ΕΑΝ ΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ ΕΝΟΣ ΠΕΛΑΤΗ ΑΠΑΙΤΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕ ΓΛΩΣΣΑ ΕΚΤΟΣ ΤΩΝ ΑΓΓΛΙΚΩΝ, ΑΠΟΤΕΛΕΙ ΕΥΘΥΝΗ ΤΟΥ ΠΕΛΑΤΗ ΝΑ ΠΑΡΕΧΕΙ ΥΠΗΡΕΣΙΕΣ ΜΕΤΑΦΡΑΣΗΣ.
- ΜΗΝ ΕΠΙΧΕΙΡΗΣΕΤΕ ΤΗΝ ΕΚΤΕΛΕΣΗ ΕΡΓΑΣΙΩΝ ΣΕΡΒΙΣ ΣΤΟΝ ΕΞΟΠΛΙΣΜΟ ΕΚΤΟΣ ΕΑΝ ΕΧΕΤΕ ΣΥΜΒΟΥΛΕΥΤΕΙ ΚΑΙ ΕΧΕΤΕ ΚΑΤΑΝΟΗΣΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ.
- ΕΑΝ ΔΕ ΛΑΒΕΤΕ ΥΠΟΨΗ ΤΗΝ ΠΡΟΕΙΔΟΠΟΙΗΣΗ ΑΥΤΗ, ΕΝΔΕΧΕΤΑΙ ΝΑ ΠΡΟΚΛΗΘΕΙ ΤΡΑΥΜΑΤΙΣΜΟΣ ΣΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ, ΣΤΟ ΧΕΙΡΙΣΤΗ Ή ΣΤΟΝ ΑΣΘΕΝΗ ΑΠΟ ΗΛΕΚΤΡΟΠΛΗΞΙΑ, ΜΗΧΑΝΙΚΟΥΣ Ή ΑΛΛΟΥΣ ΚΙΝΔΥΝΟΥΣ.

**FIGYELMEZTETÉS**

(HU)

- EZEN KARBANTARTÁSI KÉZIKÖNYV KIZÁRÓLAG ANGOL NYELVEN ÉRHETŐ EL.
- HA A VEVŐ SZOLGÁLTATÓJA ANGOLTÓL ELTÉRŐ NYELVRE TART IGÉNYT, AKKOR A VEVŐ FELELŐSSÉGE A FORDÍTÁS ELKÉSZÍTTETÉSE.
- NE PRÓBÁLJA ELKEZDENI HASZNÁLNI A BERENDEZÉST, AMÍG A KARBANTARTÁSI KÉZIKÖNYVBEN LEÍRTAKAT NEM ÉRTELMEZTÉK.
- EZEN FIGYELMEZTETÉS FIGYELMEN KÍVÜL HAGYÁSA A SZOLGÁLTATÓ, MŰKÖDTETŐ VAGY A BETEG ÁRAMÚTÉS, MECHANIKAI VAGY EGYÉB VESZÉLYHELYZET MIATTI SÉRÜLÉSÉT EREDMÉNYEZHETI.

**AÐVÖRUN****(IS)**

- ÞESSI ÞJÓNUSTUHANDBÓK ER EINGÖNGU FÁANLEG Á ENSKU.
- EF AÐ ÞJÓNUSTUVEITANDI VIÐSKIPTAMANNS ÞARFNAST ANNAS TUNGUMÁLS EN ENSKU, ER ÞAÐ SKYLD AÐ VIÐSKIPTAMANNS AÐ SKAFFA TUNGUMÁLAPJÓNUSTU.
- REYNIÐ EKKI AÐ AFGREIÐA TÆKIÐ NEMA AÐ ÞESSI ÞJÓNUSTUHANDBÓKHEFUR VERIÐ SKOÐUÐ OG SKILIN.
- BROT Á SINNA ÞESSARI AÐVÖRUN GETUR LEITT TIL MEIÐSLA Á ÞJÓNUSTUVEITANDA, STJÓRNANDA EÐA SJÚKLINGS FRÁ RAFLOSTI, VÉLRÆNU EÐA ÖÐRUM ÁHÆTTUM.

**AVVERTENZA****(IT)**

- IL PRESENTE MANUALE DI MANUTENZIONE E DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

**警告****(JA)**

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

**경고****(KO)**

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다.
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오.
- 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운전자 혹은 환자에게 위해를 가할 수 있습니다.

**BRĪDINĀJUMS  
(LV)**

- ŠĪ APKALPES ROKASGRĀMATA IR PIEEJAMA TIKAI ANGLŪ VALODĀ.
- JA KLIENTA APKALPES SNIEDZĒJAM NEPIECIEŠAMA INFORMĀCIJA CITĀ VALODĀ, NEVIS ANGLŪ, KLIENTA PIENĀKUMS IR NODROŠINĀT TULKOŠANU.
- NEVEICIET APRĪKOJUMA APKALPI BEZ APKALPES ROKASGRĀMATAS IZLASĪŠANAS UN SAPRAŠANAS.
- ŠĪ BRĪDINĀJUMA NEIEVĒROŠANA VAR RADĪT ELEKTRISKĀS STRĀVAS TRIECIENA, MEHĀNISKU VAI CITU RISKU IZRAISĪTU TRAUMU APKALPES SNIEDZĒJAM, OPERATORAM VAI PACIENTAM.

**[SPĒJIMAS  
(LT)**

- ŠIS EKSPLOATAVIMO VADOVAS YRA PRIEINAMAS TIK ANGLŲ KALBA.
- JEI KLIENTO PASLAUGŲ TIEKĒJAS REIKALAUJA VADOVO KITA KALBA – NE ANGLŲ, NUMATYTI VERTIMO PASLAUGAS YRA KLIENTO ATSAKOMYBĒ.
- NEMĖGINKITE ATLIKTI ĮRANGOS TECHNINĖS PRIEŽIŪROS, NEBENT ATSIŽVELGĖTE Į ŠĮ EKSPLOATAVIMO VADOVĄ IR JĮ SUPRATOTE.
- JEI NEATKREIPSITE DĖMESIO Į ŠĮ PERSPĖJIMĄ, GALIMI SUŽALOJIMAI DĖL ELEKTROS ŠOKO, MECHANINIŲ AR KITŲ PAVOJŲ PASLAUGŲ TIEKĖJUI, OPERATORIUI AR PACIENTUI.

**ADVARSEL  
(NO)**

- DENNE SERVICEHÅNDBOKEN FINNES BARE PÅ ENGELSK.
- HVIS KUNDENS SERVICELEVERANDØR TRENGER ET ANNET SPRÅK, ER DET KUNDENS ANSVAR Å SØRGE FOR OVERSETTELSE.
- IKKE FORSØK Å REPARERE UTSTYRET UTEN AT DENNE SERVICEHÅNDBOKEN ER LEST OG FORSTÅTT.
- MANGLENDE HENSYN TIL DENNE ADVARSELEN KAN FØRE TIL AT SERVICELEVERANDØREN, OPERATØREN ELLER PASIENTEN SKADES PÅ GRUNN AV ELEKTRISK STØT, MEKANISKE ELLER ANDRE FARER.

**OSTRZEŻENIE  
(PL)**

- NINIEJSZY PODRĘCZNIK SERWISOWY DOSTĘPNY JEST JEDYNIE W JĘZYKU ANGIELSKIM.
- JEŚLI DOSTAWCA USŁUG KLIENTA WYMAGA JĘZYKA INNEGO NIŻ ANGIELSKI, ZAPEWNIENIE USŁUGI TŁUMACZENIA JEST OBOWIĄZKIEM KLIENTA.
- NIE PRÓBOWAĆ SERWISOWAĆ WYPOSAŻENIA BEZ ZAPOZNANIA SIĘ I ZROZUMIENIA NINIEJSZEGO PODRĘCZNIKA SERWISOWEGO.
- NIEZASTOSOWANIE SIĘ DO TEGO OSTRZEŻENIA MOŻE SPOWODOWAĆ URAZY DOSTAWCY USŁUG, OPERATORA LUB PACJENTA W WYNIKU PORAŻENIA ELEKTRYCZNEGO, ZAGROŻENIA MECHANICZNEGO BĄDŹ INNEGO.



**ATENÇÃO  
(PT)**

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENDE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

**ATENȚIE  
(RO)**

- ACEST MANUAL DE SERVICE ESTE DISPONIBIL NUMAI ÎN LIMBA ENGLEZĂ.
- DACĂ UN FURNIZOR DE SERVICII PENTRU CLIEȚI NECESITĂ O ALTĂ LIMBĂ DECÂT CEA ENGLEZĂ, ESTE DE DATORIA CLIENTULUI SĂ FURNIZEZE O TRADUCERE.
- NU ÎNCERCAȚI SĂ REPARAȚI ECHIPAMENTUL DECÂT ULTERIOR CONSULTĂRII ȘI ÎNȚELEGERII ACESTUI MANUAL DE SERVICE.
- IGNORAREA ACESTUI AVERTISMENT AR PUTEA DUCE LA RĂNIREA DEPARATORULUI, OPERATORULUI SAU PACIENTULUI ÎN URMA PERICOLELOR DE ELECTROCUTARE, MECANICE SAU DE ALTĂ NATURĂ.

**ОСТОРОЖНО!  
(RU)**

- ДАННОЕ РУКОВОДСТВО ПО ОБСЛУЖИВАНИЮ ПРЕДЛАГАЕТСЯ ТОЛЬКО НА АНГЛИЙСКОМ ЯЗЫКЕ.
- ЕСЛИ СЕРВИСНОМУ ПЕРСОНАЛУ КЛИЕНТА НЕОБХОДИМО РУКОВОДСТВО НЕ НА АНГЛИЙСКОМ, А НА КАКОМ-ТО ДРУГОМ ЯЗЫКЕ, КЛИЕНТУ СЛЕДУЕТ САМОСТОЯТЕЛЬНО ОБЕСПЕЧИТЬ ПЕРЕВОД.
- ПЕРЕД ОБСЛУЖИВАНИЕМ ОБОРУДОВАНИЯ ОБЯЗАТЕЛЬНО ОБРАТИТЕСЬ К ДАННОМУ РУКОВОДСТВУ И ПОЙМИТЕ ИЗЛОЖЕННЫЕ В НЕМ СВЕДЕНИЯ.
- НЕСООБЛЮДЕНИЕ ТРЕБОВАНИЙ ДАННОГО ПРЕДУПРЕЖДЕНИЯ МОЖЕТ ПРИВЕСТИ К ТОМУ, ЧТО СПЕЦИАЛИСТ ПО ОБСЛУЖИВАНИЮ, ОПЕРАТОР ИЛИ ПАЦИЕНТ ПОЛУЧАТ УДАР ЭЛЕКТРИЧЕСКИМ ТОКОМ, МЕХАНИЧЕСКУЮ ТРАВМУ ИЛИ ДРУГОЕ ПОВРЕЖДЕНИЕ.

- UPOZORNENIE**  
**(SK)**
- TENTO NÁVOD NA OBSLUHU JE K DISPOZÍCII LEN V ANGLIČTINE.
  - AK ZÁKAZNÍKOV POSKYTOVATEĽ SLUŽIEB VYŽADUJE INÝ JAZYK AKO ANGLIČTINU, POSKYTNUTIE PREKLADATEĽSKÝCH SLUŽIEB JE ZODPOVEDNOSŤOU ZÁKAZNÍKA.
  - NEPOKÚŠAJTE SA O OBSLUHU ZARIADENIA SKÔR, AKO SI NEPREČÍTATE NÁVOD NA OBLUHU A NEPOROZUMIETE MU.
  - ZANEDBANIE TOHTO UPOZORNENIA MÔŽE VYÚSTIŤ DO ZRANENIA POSKYTOVATEĽA SLUŽIEB, OBSLUHUJÚCEJ OSOBY ALEBO PACIENTA ELEKTRICKÝM PRÚDOM, DO MECHANICKÉHO ALEBO INÉHO NEBEZPEČENSTVA.
- ATENCION**  
**(ES)**
- ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES.
  - SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION.
  - NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
  - LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.
- VARNING**  
**(SV)**
- DEN HÄR SERVICEHANDBOKEN FINNS BARA TILLGÄNGLIG PÅ ENGELSKA.
  - OM EN KUNDS SERVICETEKNIKER HAR BEHOV AV ETT ANNAT SPRÅK ÄN ENGELSKA ANSVARAR KUNDEN FÖR ATT TILLHANDAHÅLLA ÖVERSÄTTNINGSTJÄNSTER.
  - FÖRSÖK INTE UTFÖRA SERVICE PÅ UTRUSTNINGEN OM DU INTE HAR LÄST OCH FÖRSTÅR DEN HÄR SERVICEHANDBOKEN.
  - OM DU INTE TAR HÄNSYN TILL DEN HÄR VARNINGEN KAN DET RESULTERA I SKADOR PÅ SERVICETEKNIKERN, OPERATÖREN ELLER PATIENTEN TILL FÖLJD AV ELEKTRISKA STÖTAR, MEKANISKA FAROR ELLER ANDRA FAROR.

**DİKKAT****(TR)**

- BU SERVİS KILAVUZUNUN SADECE İNGİLİZCESİ MEVCUTTUR.
- EĞER MÜŞTERİ TEKNİSYENİ BU KILAVUZU İNGİLİZCE DIŞINDA BİR BAŞKA LİSANDAN TALEP EDERSE, BUNU TERCÜME ETTİRMEK MÜŞTERİYE DÜŞER.
- SERVİS KILAVUZUNU OKUYUP ANLAMADAN EKİPMANLARA MÜDAHALE ETMEYİNİZ.
- BU UYARIYA UYULMAMASI, ELEKTRİK, MEKANİK VEYA DİĞER TEHLİKELERDEN DOLAYI TEKNİSYEN, OPERATÖR VEYA HASTANIN YARALANMASINA YOL AÇABİLİR.

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## Revision History

| Rev. | Date         | Primary Reason(s) for Change |
|------|--------------|------------------------------|
| 1    | May 28, 2008 | Initial Release              |

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# Chapter 1 Preinstallation

## 1 Unpacking the High Order Shim Hardware

### 1.1 Personnel Requirements

| Required Persons | Preliminary Reqs | Procedure          | Finalization   |
|------------------|------------------|--------------------|----------------|
| 1                | Not Applicable   | 2 hours to install | Not Applicable |

**NOTE:** It will take an additional 2 hours to run the functional check and to build a shim calibration file, so plan accordingly.

### 1.2 Overview

The commercial catalog M3000PP, High Order Shim Option, is compatible with and can be installed on the following product configurations:

- Signa Excite 1.5T
- Signa HD 1.5T
- Signa HDx 1.5T and 3.0T

Verify that all parts are present in the shipment (see [Table 1-1](#) ).

**Table 1-1: Catalog M3000PP Contents List**

| Item | Qty | Part Number | Description                                                                  |
|------|-----|-------------|------------------------------------------------------------------------------|
| 1    | 1   | 2352973     | Shim Filter Box Assembly                                                     |
| 2    | 1   | 2352972     | Shim Power Supply Enclosure Assembly (Y)                                     |
| 3    | 1   | 2303317     | CAN Termination (Male) (Y)                                                   |
| 4    | 1   | 2363363-4   | Cable; From Shim PS J4 to TAC I/O J119                                       |
| 5    | 1   | 2374156-2   | Cable; From Shim PS J3 To TAC I/O J126                                       |
| 6    | 1   | 2384104-2   | Cable; Run# 1152; From PP1-A5-P1 To MG2A12A3                                 |
| 7    | 1   | 5113502-5   | Cable; Run# 1233 From TAC I/O J116 To PP1-PHPS-J155                          |
| 8    | 1   | 5117890     | Cable; From Shim PS J5 to TAC I/O 116                                        |
| 9    | 1   | 5118283     | Cable; Run# 611; From TAC I/O J126 to PP1-A5-P1                              |
| 10   | 1   | 2322427     | Cable; Power Cord NEMA Male To IEC320F 7ft long                              |
| 11   | 4   | 2279742-2   | ID 0.2040 OD 0.5630 THK .0478STL F70B5A (for securing the Shim Power Supply) |
| 12   | 4   | 2279742-2   | Screw; Pan Head 5 mm x 16 mm (for securing the Shim Power Supply)            |

**Table 1-1: Catalog M3000PP Contents List (cont'd)**

|    |   |                   |                                                                                                                                                        |
|----|---|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | 4 | 46-170012P35      | 6-32, 1/4 inch Hex Keps Nut, 3/32 Thick Overall Height 7/64; Washer OD 0/32 Washer Material Thk .013; CAD PL Keps (for mounting connector to I/O J126) |
| 14 | 4 | 46-208560P28      | Screw; Pan Head; 6-32 X .500 inch; Zinc Plated (for mounting connector to I/O J126)                                                                    |
| 15 | 6 | 46-221417P2       | Female Screw lock for Sub-d Connectors 4-40 Thread; 0.503 inch Overall Length; Individual Piece Without Nut Or Washer                                  |
| 16 | 1 | Direction 5314688 | Service Manual                                                                                                                                         |

### 1.3 Product Locator

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

At this time, for U.S. ONLY, the preferred method of submitting information is the FE Site Verification Web Site at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – <http://supportcentral.ge.com/15563>

One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

1. Verify that serial and model number on each rating plate matches the installation card numbers before removing installation card.

**NOTE:** There may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

2. Enter the model and serial number information into the GIB database.

## Chapter 2 Safety

### 1 Lock Out/Tag Out

#### 1.1 Safety



#### **⚠ DANGER**

**ELECTROCUTION HAZARD.  
HIGH VOLTAGE PRESENT.  
USE PROPER LOCKOUT / TAGOUT PROCEDURES BEFORE INSTALLING THE  
COMPONENTS.**



#### **NOTICE**

**Follow ALL required safety and PPE procedures customary for your organization when working on this product.**

#### 1.2 Required Contitions

Before starting any hardware installation, the MRI system must be powered down and LOTO procedures executed.

| Condition                            | Reference                                                                                                             | Eff |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----|
| System PDU locked out and tagged out | Refer to the Lockout / Tagout for Gradient Amplifier / PDU Prcedure for the specific Signa system you are working on. |     |

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## Chapter 3 Installation

### 1 Shim Filter Installation

#### 1.1 Personnel Requirements

| Required Persons | Preliminary Reqs | Procedure | Finalization   |
|------------------|------------------|-----------|----------------|
| 1                | Not Applicable   | 30 mins   | Not Applicable |

#### 1.2 Preliminary Requirements

##### 1.2.1 Tools and Test Equipment

| Item                                           | Qty | Effectivity | Part#          | Manufacturer |
|------------------------------------------------|-----|-------------|----------------|--------------|
| 9/32 inch socket and driver or open-end wrench | 1   | -           | Field supplied |              |

##### 1.2.2 Consumables

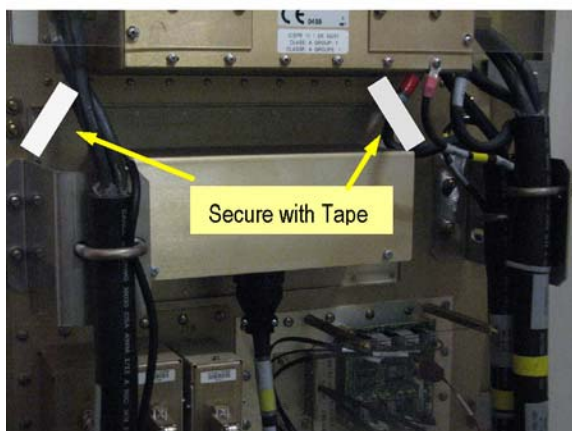
| Item         | Qty | Effectivity | Part#          | Manufacturer |
|--------------|-----|-------------|----------------|--------------|
| Masking tape | 1   | -           | Field supplied |              |

#### 1.3 Procedure

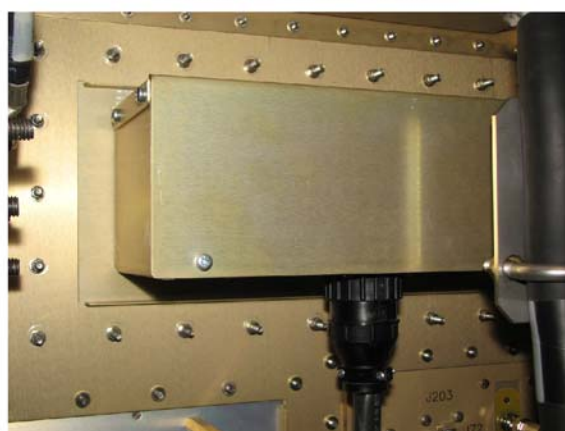
1. Remove the blank panel located below the Gradient Filter on the Penetration Panel.
2. Remove the nuts and washers attached to the studs on the Shim Filter Assembly.
3. Install the Shim Filter Assembly, p/n 2352973, into the opening on the Penetration Panel from the Equipment Room side of the panel. See [Illustration 3-1](#) , Equipment Room View.

**Illustration 3-1: Penetration Panel Views**

Equipment Room View



Magnet Room View



4. Apply strips of masking tape across the corners of the Shim Filter assembly to temporarily hold it in place.
5. Secure the Shim Filter assembly using the nuts and washers, which were previously removed in step 2. See [Illustration 3-1](#) , Magnet Room View.
6. Make sure all the nuts are tightened to enable a good RF seal on the panel.

## **1.4 Finalization**

No finalization steps.

## 2 Shim Power Supply Installation

### 2.1 Personnel Requirements

| Required Persons | Preliminary Reqs | Procedure | Finalization   |
|------------------|------------------|-----------|----------------|
| 1                | Not Applicable   | 30 mins   | Not Applicable |

### 2.2 Preliminary Requirements

#### 2.2.1 Tools and Test Equipment

| Item                   | Qty | Effectivity | Part#                     | Manufacturer |
|------------------------|-----|-------------|---------------------------|--------------|
| #2 Philips Screwdriver | 1   | -           | 5109894-3 or equivalent   |              |
| Allen key 5/32 inch    | 1   | -           | 5109894\2-8 or equivalent |              |

#### 2.2.2 Safety

**NOTE:** The Shim Power Supply weighs 36 lbs (16.4 Kgs). Use proper lifting techniques when handling the Shim Power Supply.

### 2.3 Procedure

1. Remove the Front Cover from the TAC Cabinet using a 5/32" Allen key to open the cover lock.
2. Install the Shim Power Supply onto the Slide Rails located above the Gradient Switch Assembly. See [Illustration 3-2](#).

**Illustration 3-2: Shim Power Supply Front View**



3. Secure the Shim Power Supply to the vertical rails using 4 flat washers and 4 pan head Philips screws.

## **2.4 Finalization**

No finalization steps.



### 3 Cable Installation

#### 3.1 Personnel Requirements

| Required Persons | Preliminary Reqs | Procedure | Finalization |
|------------------|------------------|-----------|--------------|
| 1                | Not Applicable   | 1 hr      | 2 hrs        |

#### 3.2 Preliminary Requirements

##### 3.2.1 Tools and Test Equipment

| Item                             | Qty | Effectivity | Part#                   | Manufacturer |
|----------------------------------|-----|-------------|-------------------------|--------------|
| 1/ inch Flat Blade Screwdriver   | 1   | -           | 5109895-4 or equivalent |              |
| 3/16 inch Flat Blade Screwdriver |     | -           | 5109895-7 or equivalent |              |
| 3/16 inch Nut Driver             | 1   | -           | Field Supplied          |              |

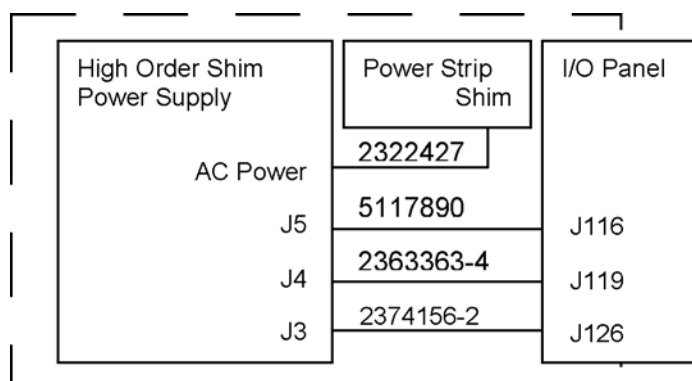
##### 3.2.2 Consumables

| Item       | Qty | Effectivity | Part#          | Manufacturer |
|------------|-----|-------------|----------------|--------------|
| Cable Ties | 20  | -           | Field supplied | -            |

#### 3.3 Procedure

1. Install the internal cabinet cables between the Shim Power Supply and the I/O Panel as shown in wiring diagram [Illustration 3-3](#) .

**Illustration 3-3: TAC Cabinet Wiring Diagram**



2. Attach the bulkhead connector of each cable to the I/O Panel as defined in [Illustration 3-3](#) .

**NOTE:** For mounting connector to J126 use the following mounting hardware:

- 4 each; 46-208560P28; Screw; Pan Head; 6-32 X .500 inch
- 4 each; 46-170012P35; 6-32, 1/4 inch Hex Keys Nut

**NOTE:** For mounting connectors to J116 and J119 use the following mounting hardware:

2 each; 46-221417P2 – Female screw lock for Sub-d Connectors 4-40 Thread

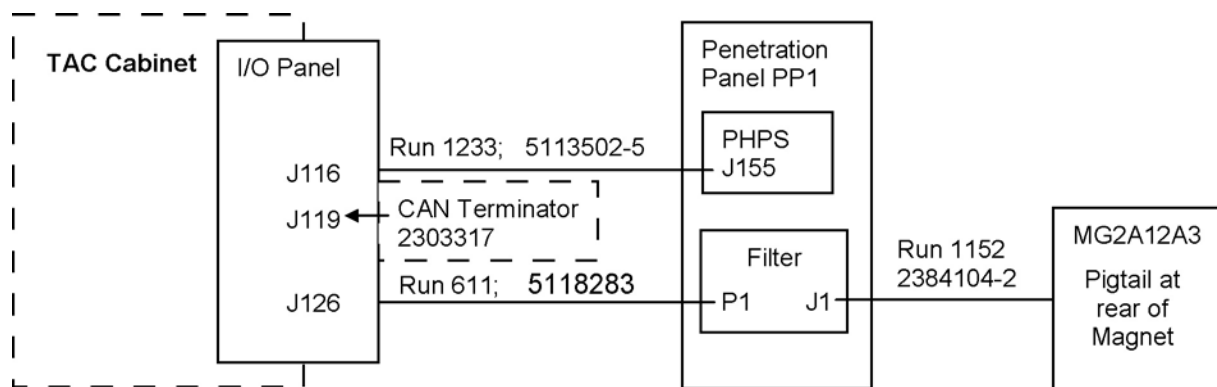
3. Route and secure the cables to the vertical rails inside the cabinet as shown in [Illustration 3-4](#) using cable ties.

**Illustration 3-4: Internal Cabinet Cable Routing**



4. Install the external cabinet cables between the TAC I/O Panel and the Penetration Panel as defined in [Illustration 3-5](#).

**Illustration 3-5: External Cabinet Cable Diagram**



### **3.4 Finalization**

1. Refer to the service methods CD for the Signa system you are working on and perform the following tasks:
  - Remove LOTO for Gradient Amplifier / PDU
  - Power Up the System
  - Perform High Order Shim Supply Functional Check
  - Perform HOShim Calibration Reference Map Creation and Procedures
  - Perform HOShim Calibration Reference Map Creation and Procedures

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# Chapter 4 Parts

## 1 Replacement Parts

| Item                       | Qty | Effectivity | Part Number | Manufacturer |
|----------------------------|-----|-------------|-------------|--------------|
| Shim Filter Assembly       | 1   |             | 2352973     |              |
| Shim Power Supply Assembly | 1   |             | 2352972     |              |
| CAN Terminator             | 1   |             | 2303317     |              |

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